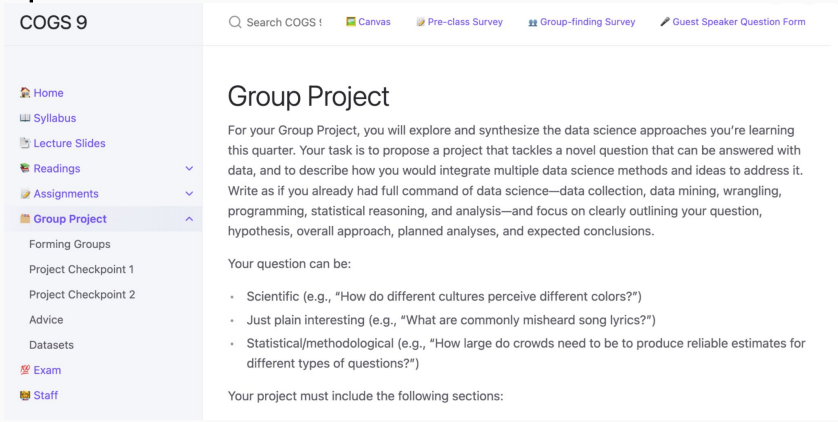
COGS 9 Section

TA: Emily Cheng

Friday, October 23

Group Project Checkpoint 1

- Due Weds, October 28th @ 11:59pm on Gradescope!
- You should have received an email from Lindsey and I if you filled out a Google form.
- Checkpoint 1 template on course website



Today we'll:

- Review the Checkpoint 1 requirements
- Discuss how to develop a good data science question
- Clarify expectations for datasets and ethical considerations
- Give you time to work with your group
- Answer questions about your projects

What makes a good data science question?

A good data science question should be:

- Clear and understandable
- Answerable using data
- Specific enough to identify relevant variables
- Realistic given the time and resources available

Example

Too broad: "Does social media affect students?"

Better: "Is daily social media use associated with sleep duration among college students?"

Remember: Your question can evolve as you explore available data!

Hypothesis: What do you expect to find?

Example:

"We hypothesize that college students who spend more time on social media tend to report shorter sleep durations."

Your hypothesis should:

- Clearly state your expected result
- Relate directly to your research question
- Include a brief justification

What data would you need to answer your question?

describe your *ideal dataset*:

- What are your observations? (Who or what are you studying?)
- What variables do you need?
- Approximately how many observations would you want?

Option A: Find an existing dataset

- Provide a link to the dataset.
- Describe its observations and variables.
- Explain how it differs from your ideal dataset.

Option B: Propose collecting your own data

- Explain what information you would collect.
- Describe how you would collect it.

Ethical Considerations

Stage	Questions to consider
Data Collection	Do participants consent? Is the sample representative?
Data Storage	How will information be secured? When will it be deleted?
Data Analysis	Are there biases or missing perspectives? Are results represented honestly?
Modeling	Could predictions discriminate against certain groups? Are limitations explained?
Deployment	Could the results cause unintended harm? How would problems be addressed?

Attendance Webclicker

webclicker.web.app



Code: OMFZUC

Different from lecture code!!

If you have issues using webclicker, please let us know before you leave the section!

How was Midterm 1?

A. 😭 B. 😎 C. 🤩 D. 😀 E. 🫠

Work with your group on Checkpoint 1!

During this time, try to:

1. Agree on a research question.
2. Write a preliminary hypothesis.
3. Identify your ideal dataset and necessary variables.
4. Find a potential dataset or develop a data collection plan.
5. Start discussing ethical considerations.

Questions?

Come talk to me if you don’t have a group yet